EXAMPLE: POOL CHEMISTRY MONITORING

Electronic pool chemistry monitors cost thousands of dollars, but simple test kits provide accurate measurements for under $50. Here's how I will integrate manual testing into my digital twin:

1. *Standardized process*. To ensure consistency, I manually test the water at the same time each day, from the same location, following an identical procedure.
2. *Digital recording*. I built a simple mobile interface that allows me to enter readings immediately after taking measurements.
3. *Contextual data*. The manual process lets me record qualitative observations like water clarity and debris level that sensors would miss entirely.

This approach will reveal patterns that would be invisible without systematic collection, such as a rise in pH on hot sunny days, changes in chemistry after heavy rainfall and will allow me to optimize the runtime of the filter seasonally to reduce electricity usage. Listing 3.3 shows how easily you can create an interface to capture manual data readings and transmit them to an MQTT broker for processing.